

Social Network Analysis of Organizational Structure

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Abstract

This project investigates whether structural patterns in the Enron social network arise from intentional organization or random processes. Using Neighborhood Degree Sequences (NDS), statistical testing, and PCA, we quantify structural properties and validate their significance. Results show statistically significant deviations from randomness and strong dimensional compressibility, indicating meaningful hierarchical organization.

Contents

1	Problem Statement	2
2	Methodology and Quantitative Findings	2
2.1	Neighborhood Degree Sequences (NDS)	2
2.2	Structural Metrics	2
2.2.1	Node Heterogeneity (V_n)	2
2.2.2	Neighborhood Similarity (S)	2
2.2.3	Hierarchical Complexity (Ω)	3
2.2.4	Network Organization (R)	3
2.3	Configuration Model Comparison	3
2.4	Statistical Significance	3
2.5	Principal Component Analysis (PCA)	4
2.6	Influence Scoring Model	4
3	Key Insights	4
4	Conclusion	5

1 Problem Statement

Determine whether observed network structure is:

- A byproduct of degree distribution, or
- Evidence of intentional organizational hierarchy

2 Methodology and Quantitative Findings

2.1 Neighborhood Degree Sequences (NDS)

Each node is represented by a 20-dimensional vector of the highest neighbor degrees.

Quantitative Insight:

- Fixed dimension = 20
- Captures local topology in a comparable vector space

Interpretation: This converts graph structure into a numerical representation where similarity and variance can be measured rigorously.

2.2 Structural Metrics

2.2.1 Node Heterogeneity (V_n)

$$V_n = 24931.98$$

Meaning:

- High variance in neighbor degrees
- Indicates nodes are connected to both highly connected and weakly connected nodes

Implication: The network is **not homogeneous**. Presence of hubs and peripheral nodes suggests hierarchical structure.

2.2.2 Neighborhood Similarity (S)

$$S = 0.5603$$

Meaning:

- 56% of nodes share similar neighborhood structures

Implication:

- Significant repetition of structural patterns
- Suggests roles or classes of nodes (e.g., managers, employees)

2.2.3 Hierarchical Complexity (Ω)

$$\Omega = 0.02229$$

Meaning:

- Low but non-zero variation among nodes of same degree

Implication: Even nodes with identical degree differ structurally $\rightarrow$ hierarchy is not purely degree-driven.

2.2.4 Network Organization (R)

$$R = 145204.39$$

Meaning:

- Large deviation from random expectation

Implication: Strong evidence of **non-random global organization**

2.3 Configuration Model Comparison

Metric	Real	CM Mean	Ratio (Real/CM)
V_n	24931.98	15024.47	1.66
S	0.5603	0.3437	1.63
Ω	0.02229	0.00671	3.32
R	145204.39	31112.81	4.67

Key Quantitative Observations:

- R is **4.67 $\times$ larger** than random $\rightarrow$ strong structural organization
- Ω is **3.32 $\times$ larger** $\rightarrow$ hierarchy beyond degree effects
- S is **1.63 $\times$ larger** $\rightarrow$ repeated structural motifs

2.4 Statistical Significance

$$p\text{-value} = 1.95 \times 10^{-3}$$

Interpretation:

- $p < 0.05 \rightarrow$ reject null hypothesis
- Probability of observing this difference by chance $< 0.2\%$

Conclusion: The network contains **statistically significant structure beyond randomness**

2.5 Principal Component Analysis (PCA)

Result:

- PC1 explains $\sim 80\%$ variance
- Remaining 19 dimensions explain only $\sim 20\%$

Quantitative Interpretation:

- Dimensionality reduction: $20 \rightarrow 1$ with minimal information loss
- Compression ratio = 20:1

Implication: Network structure is governed by a **dominant latent factor**, likely representing influence or connectivity hierarchy.

2.6 Influence Scoring Model

$$\text{Influence} = \alpha \cdot \text{Degree} + (1 - \alpha) \cdot \text{PC1}$$

Quantitative Findings:

Node	Degree	PC1	Influence Score
273	1367	15.63	4.65
136	1026	16.34	4.64
1028	1244	15.78	4.63

Interpretation:

- High degree alone is insufficient
- Nodes with slightly lower degree but higher PC1 remain competitive

Implication:

- Influence depends on:
 - Quantity of connections (degree)
 - Quality of connections (neighbor importance)

3 Key Insights

- Network organization is **4–5 $\times$ stronger than random expectation**
- More than **50% of nodes share structural patterns**
- **80% of structural variance** is explained by a single component
- Influence is a **hybrid property**, not purely degree-based

4 Conclusion

The Enron network exhibits:

- Strong non-random organization
- Detectable hierarchical structure
- High compressibility in structural representation

Quantitatively, all metrics significantly exceed random baselines, confirming that observed patterns arise from intentional or functional organization rather than chance.